



EduBuddy - AI Learning Assistant Solution for SDG 4: Quality Education

SUBJECT: LENOVO LEAP AI CAPSTONE PROJECT (MILESTONE 8)

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TOPIC: AI BASED CHATBOT

COURSE/BATCH: AI INTERNSHIP/JUNE 2026

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1. SDG & Problem Statement

 **Selected SDG:** Goal 4 - Quality Education

 **Why this SDG ?:** Many students, especially in rural areas, don't have access to personal tutors or quick doubt-solving support. They get stuck on concepts and have no one to ask.

 **Problem Statement:**

1. **What:** Students struggle to understand concepts because they have no one to ask when stuck
2. **Where:** Rural and semi-urban areas with limited access to coaching
3. **Who affected:** School and college students
4. **Why serious:** Without guidance, students lose interest or fall behind
5. **If not solved:** Education gap widens; more students rely on rote learning

2. Solution Description

 **Project Name:** EduBuddy - AI Learning Assistant

 **How it works:**

1. Student types a question in the chat
2. AI understands the question and explains the concept in simple language
3. Student can ask follow-up questions
4. Chat history is maintained for context

 **Who will use it ?:** Students from Class 8 to college level

 **How it helps ?:** Provides instant, free, personalized learning support 24/7

3. Technology Stack

Component	Technology used
Frontend	Streamlit(via Python)
AI Model/API	Groq API
Programming Language	Python
Deployment	Hugging Face Spaces
Version Control	GitHub

4. Results (Code Snippet 1)



```
1  # EduBuddy - AI Learning Assistant (Powered by Groq)
2  # SDG 4: Quality Education
3
4  import streamlit as st
5  from groq import Groq
6  from dotenv import load_dotenv
7  import os
8
9  # Load API key from .env file
10 load_dotenv()
11
12 # Get API key
13 API_KEY = os.getenv("GROQ_API_KEY")
14
15 # Check if API key exists
16 if not API_KEY:
17     st.error("⚠️ GROQ_API_KEY not found. Please add it to your .env file.")
18     st.stop()
19
20 # Initialize Groq client
21 try:
```

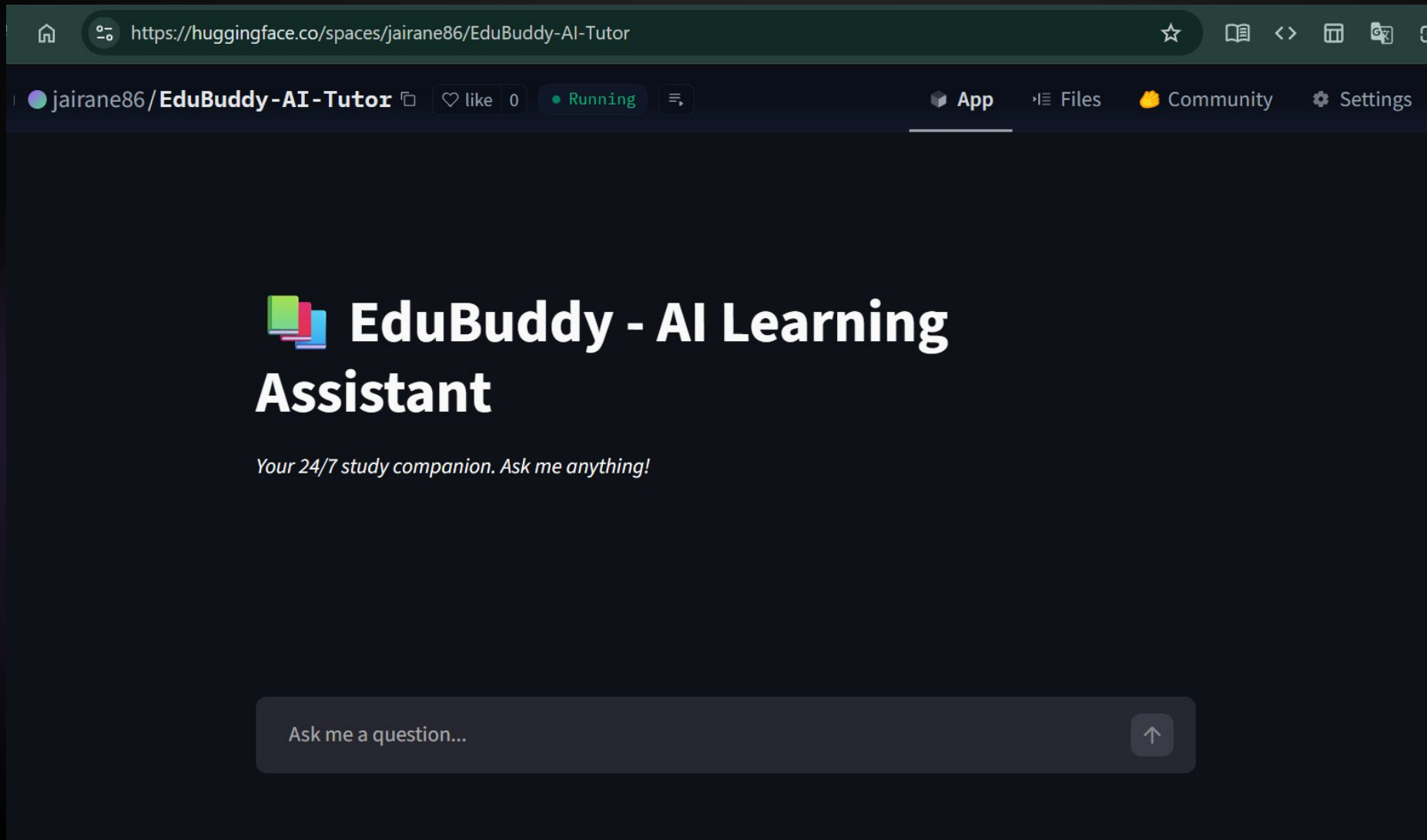

4. Results (Code Snippet 2)

```
22     client = Groq(api_key=API_KEY)
23 except Exception as e:
24     st.error(f"Failed to initialize Groq client: {e}")
25     st.stop()
26
27 # Page setup
28 st.set_page_config(page_title="EduBuddy - AI Study Buddy", page_icon="📖")
29
30 # Title
31 st.title("📖 EduBuddy - AI Learning Assistant")
32 st.markdown("*Your 24/7 study companion. Ask me anything!*")
33
34 # Initialize chat history
35 if "messages" not in st.session_state:
36     st.session_state.messages = []
37
38 # Display chat history
39 for message in st.session_state.messages:
40     with st.chat_message(message["role"]):
41         st.markdown(message["content"])
42
43 # Chat input
44 if prompt := st.chat_input("Ask me a question..."):
45     # Add user message to history
46     st.session_state.messages.append({"role": "user", "content": prompt})
47     with st.chat_message("user"):
```

4. Results (Code Snippet 3)

```
48         st.markdown(prompt)
49
50     # Get AI response
51     with st.chat_message("assistant"):
52         with st.spinner("Thinking..."):
53             try:
54                 # Prepare messages for Groq
55                 messages = []
56                 for m in st.session_state.messages:
57                     messages.append({"role": m["role"], "content": m["content"]})
58
59                 # Call Groq API
60                 chat_completion = client.chat.completions.create(
61                     messages=messages,
62                     model="llama-3.1-8b-instant", # Updated model name
63                 )
64                 reply = chat_completion.choices[0].message.content
65                 st.markdown(reply)
66                 st.session_state.messages.append({"role": "assistant", "content": reply})
67             except Exception as e:
68                 st.error(f"Error: {e}")
69                 st.info("Try asking a simpler question or check your API key.")
```


4. Results (AI Chatbot - EduBuddy Main Interface)



4. Results (AI Chatbot - EduBuddy Response 1)

 Spaces |  jairane86/**EduBuddy-AI-Tutor**  like 0  Running  Logs

App Files Community Settings

 **EduBuddy - AI Learning Assistant**

Your 24/7 study companion. Ask me anything!

 Explain photosynthesis in short

 Photosynthesis is the process by which plants, algae, and some bacteria convert light energy from the sun into chemical energy in the form of glucose. Here's a simplified overview:

1. **Light absorption:** Plants absorb light energy from the sun through specialized pigments such as chlorophyll.
2. **Water absorption:** Plants absorb water from the soil through their roots.
3. **Carbon dioxide absorption:** Plants absorb carbon dioxide from the air through small openings called stomata.
4. **Conversion:** The absorbed light energy, water, and carbon dioxide are converted into glucose (a type of sugar) and oxygen.
5. **Release:** The plants release oxygen into the air and use glucose for growth, development, and energy.

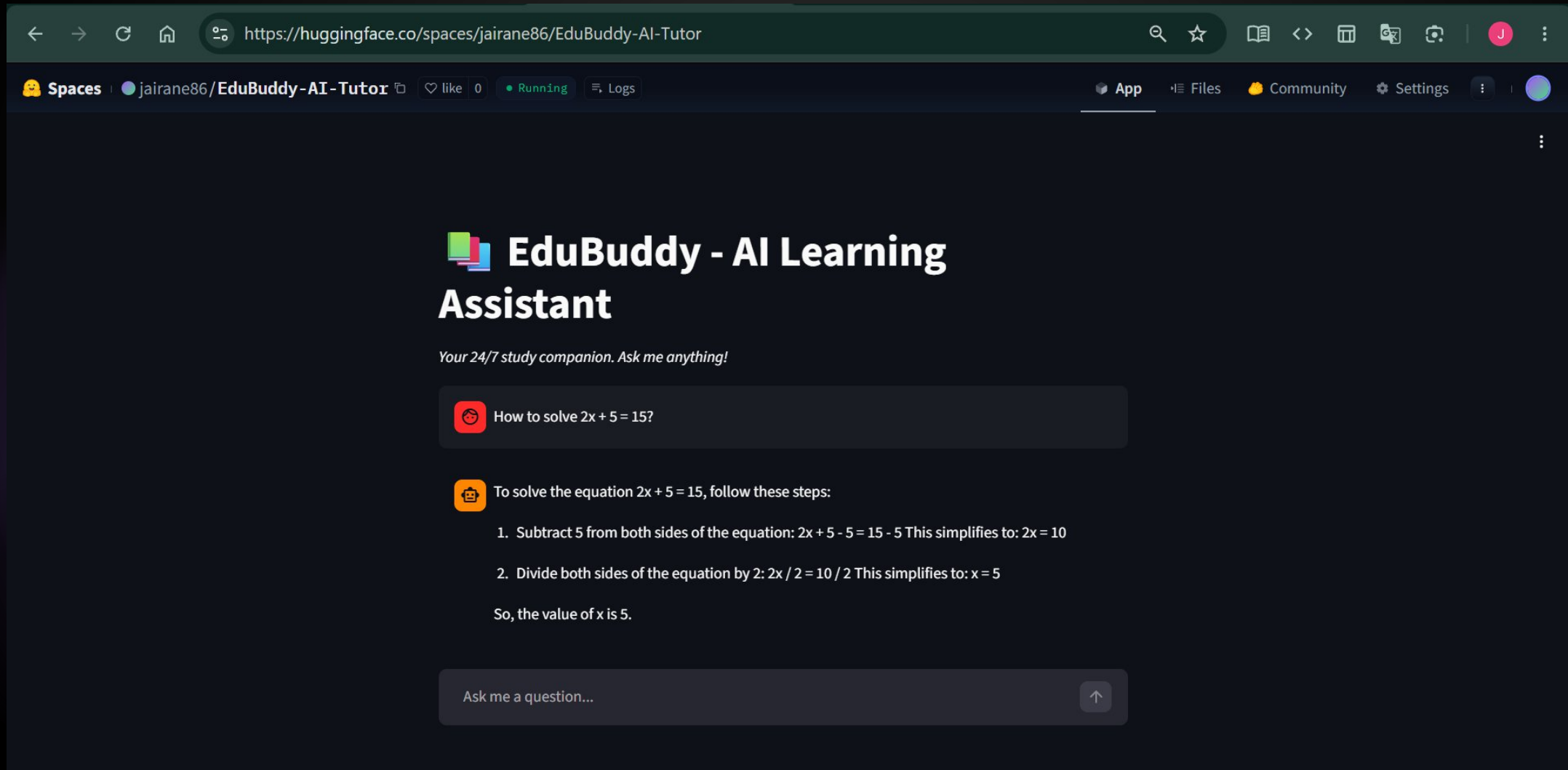
The overall equation for photosynthesis is:

$$6 \text{ CO}_2 + 6 \text{ H}_2\text{O} + \text{light energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 \text{ (glucose)} + 6 \text{ O}_2$$

This essential process supports life on Earth by providing energy for plants and oxygen for animals.

Ask me a question... 

4. Results (AI Chatbot - EduBuddy Response 2)



The screenshot shows a web browser displaying the Hugging Face Spaces page for 'jairane86/EduBuddy-AI-Tutor'. The browser's address bar shows the URL 'https://huggingface.co/spaces/jairane86/EduBuddy-AI-Tutor'. The page header includes navigation links for 'Spaces', 'jairane86/EduBuddy-AI-Tutor', 'like 0', 'Running', and 'Logs'. The main content area features the title 'EduBuddy - AI Learning Assistant' with a subtitle 'Your 24/7 study companion. Ask me anything!'. A user input box contains the question 'How to solve $2x + 5 = 15$?'. The assistant's response is displayed below, starting with 'To solve the equation $2x + 5 = 15$, follow these steps:' followed by a numbered list of two steps. The first step is '1. Subtract 5 from both sides of the equation: $2x + 5 - 5 = 15 - 5$ This simplifies to: $2x = 10$ '. The second step is '2. Divide both sides of the equation by 2: $2x / 2 = 10 / 2$ This simplifies to: $x = 5$ '. The response concludes with 'So, the value of x is 5.' At the bottom, there is an input field with the placeholder text 'Ask me a question...' and a send button.

https://huggingface.co/spaces/jairane86/EduBuddy-AI-Tutor

Spaces | jairane86/EduBuddy-AI-Tutor | like 0 | Running | Logs

App | Files | Community | Settings

EduBuddy - AI Learning Assistant

Your 24/7 study companion. Ask me anything!

How to solve $2x + 5 = 15$?

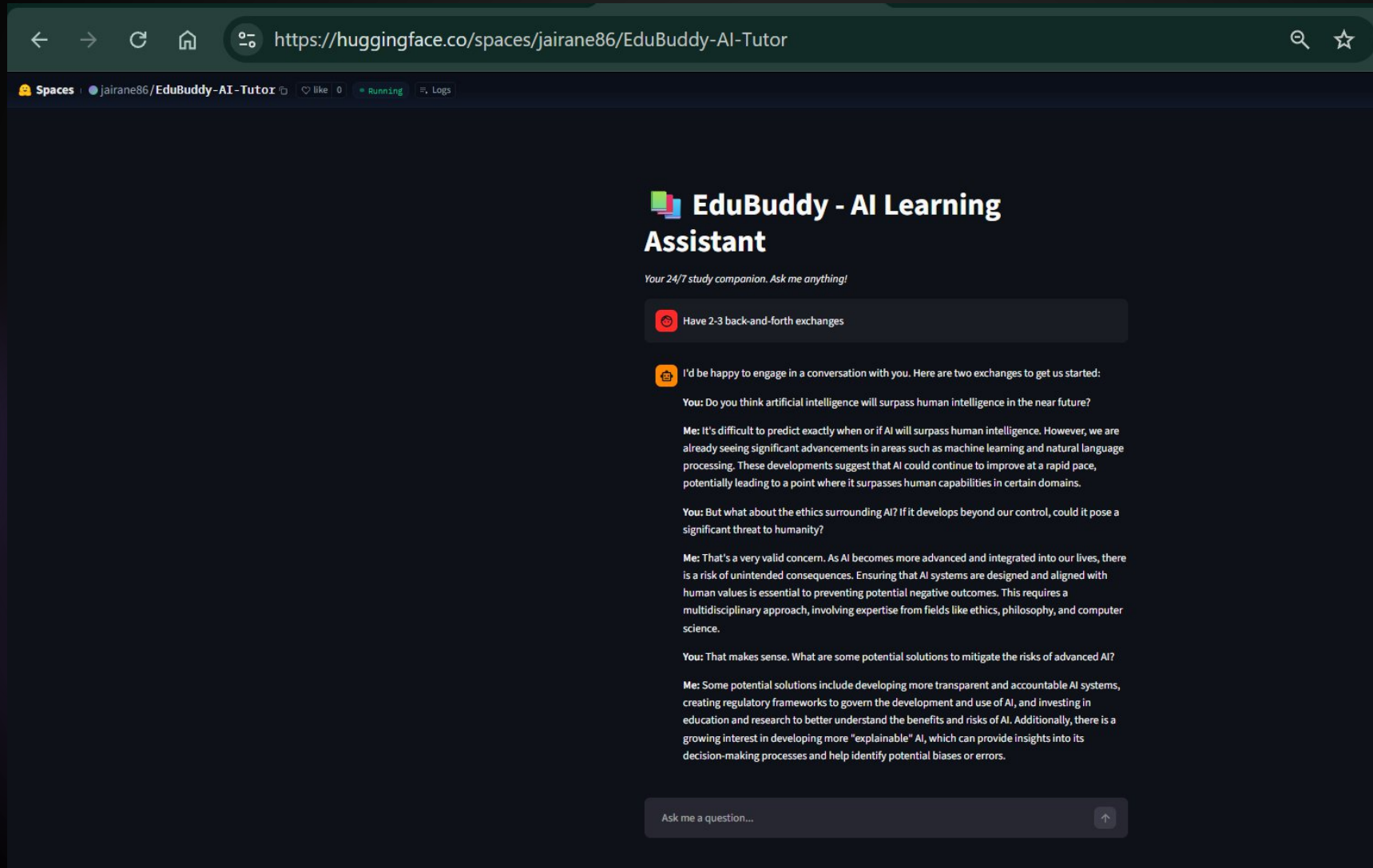
To solve the equation $2x + 5 = 15$, follow these steps:

1. Subtract 5 from both sides of the equation: $2x + 5 - 5 = 15 - 5$ This simplifies to: $2x = 10$
2. Divide both sides of the equation by 2: $2x / 2 = 10 / 2$ This simplifies to: $x = 5$

So, the value of x is 5.

Ask me a question...

4. Results (AI Chatbot - EduBuddy Response 3)



The screenshot shows a web browser window displaying the Hugging Face Spaces page for 'jairane86/EduBuddy-AI-Tutor'. The browser's address bar shows the URL 'https://huggingface.co/spaces/jairane86/EduBuddy-AI-Tutor'. The page header includes the 'Spaces' logo, the name 'jairane86/EduBuddy-AI-Tutor', a 'like' button with a count of 0, a 'Running' status indicator, and a 'Logs' link.

The main content area features the title 'EduBuddy - AI Learning Assistant' with a small icon of a book and a pencil. Below the title is a subtitle: 'Your 24/7 study companion. Ask me anything!'. A chat history section shows a previous exchange: 'Have 2-3 back-and-forth exchanges'.

The current chat session begins with a message from the AI: 'I'd be happy to engage in a conversation with you. Here are two exchanges to get us started:'. This is followed by a user question: 'You: Do you think artificial intelligence will surpass human intelligence in the near future?'. The AI responds: 'Me: It's difficult to predict exactly when or if AI will surpass human intelligence. However, we are already seeing significant advancements in areas such as machine learning and natural language processing. These developments suggest that AI could continue to improve at a rapid pace, potentially leading to a point where it surpasses human capabilities in certain domains.'.

The user asks another question: 'You: But what about the ethics surrounding AI? If it develops beyond our control, could it pose a significant threat to humanity?'. The AI responds: 'Me: That's a very valid concern. As AI becomes more advanced and integrated into our lives, there is a risk of unintended consequences. Ensuring that AI systems are designed and aligned with human values is essential to preventing potential negative outcomes. This requires a multidisciplinary approach, involving expertise from fields like ethics, philosophy, and computer science.'.

The user asks a third question: 'You: That makes sense. What are some potential solutions to mitigate the risks of advanced AI?'. The AI responds: 'Me: Some potential solutions include developing more transparent and accountable AI systems, creating regulatory frameworks to govern the development and use of AI, and investing in education and research to better understand the benefits and risks of AI. Additionally, there is a growing interest in developing more "explainable" AI, which can provide insights into its decision-making processes and help identify potential biases or errors.'.

At the bottom of the page, there is a text input field labeled 'Ask me a question...' and a button with an upward arrow icon.

5. Project Links

 Hugging Face Deployed (Live Demo):

<https://huggingface.co/spaces/jairane86/EduBuddy-AI-Tutor>

 Github: <https://github.com/ranejai954/edubuddy-ai-chatbot>

6. Future Scope

Possible improvements for the future:

1. **Voice input** - Allow students to ask questions by speaking
2. **Multiple languages** - Support Hindi, Marathi, and other Indian languages
3. **Mobile app** - Create Android/iOS version for easier access
4. **Quiz generation** - AI can test students after teaching a topic
5. **Progress tracking** - Save chat history and track learning progress

7. Conclusion

EduBuddy successfully demonstrates how AI can help achieve SDG 4 (Quality Education) by providing free, 24/7 personalized learning support to students who lack access to tutors. The chatbot uses Groq's Llama 3 model to answer questions in simple, easy-to-understand language. This project shows that AI can be a powerful tool for making education more accessible and equitable.

THANKS
FOR WATCHING